

Table 1: Long Fei FULL hexapole — Method [J] joint 6-coil fit (fitting region: cube $\pm 25 \mu\text{m}$, full FEM mesh)

$$\text{Error} = \frac{1}{N} \sum_{i=1}^N \frac{|\mathbf{B}_{model} - \mathbf{B}_{FEM}|_i}{|\mathbf{B}_{FEM}|_i} \cdot 100$$

Long Fei hexapole (Error = 0.29%)			
Pole	$ \ell $ [μm]	Tip→Charge [μm]	R_a [A/Wb]
P1	7939.8	7439.8	1.18×10^7
P2	6308.9	5808.9	1.25×10^7
P3	7008.9	6508.9	1.62×10^7
P4	5505.7	5005.7	9.36×10^6
P5	6711.7	6211.7	1.71×10^7
P6	9780.0	9280.0	8.51×10^6
Average	7209.2	6709.2	1.26×10^7

Tip→Origin = 500 μm

Tip→Charge = $|\ell| - 500$

$N_c = 70$ (70 turns per coil, Long Fei dissertation hexapole).